

**Problem 3:** Calculate the missing frequencies from the following frequency distribution of 200 observations with A.M. 1.46.

$x: 0 \quad 1 \quad 2 \quad 3 \quad 4 \quad 5$

$f: 46 \quad -- \quad -- \quad 25 \quad 10 \quad 5$

### Solution

Let  $x$  and  $y$  are the missing frequencies of 1 and 2 respectively.

Given total frequency is 200.

Hence  $46 + x + y + 25 + 10 + 5 = 200$

That is,  $x + y = 114 \quad \text{---- (1)}$

Also given  $A.M. = 1.46$

$$\Rightarrow \frac{1}{200} [0 \times 46 + 1 \times x + 2 \times y + 3 \times 25 + 4 \times 10 + 5 \times 5] = 1.46$$

$$\Rightarrow x + 2y + 140 = 292 \quad \text{i.e. } x + y = 152 \quad \text{---- (2)}$$

Solving (1) and (2), we get the missing frequencies  $x = 76$  and  $y = 38$ .

### Merits and Demerits of Arithmetic Mean

#### Merits:

1. It is rigidly defined
2. It is easy to understand and easy to calculate
3. It is based upon all the observations
4. It is amenable to further algebraic treatment. The means of the composite series in terms of the means and sizes of the component series is given by

$$\bar{x} = \frac{1}{n} \sum_{i=1}^k n_i \bar{x}_i$$

$\therefore \left(1 + \frac{1}{r} + \frac{1}{r^2} + \dots + \frac{1}{r^{n-1}}\right)$  is sum to  $n$  terms of a G.P.

with first terms 1 and common ratio  $\frac{1}{r}$ , which is equal to

$$1 \left(1 - \frac{1}{r^n}\right) / \left(1 - \frac{1}{r}\right)$$

Here,  $A.H = \frac{a(1-r^n)}{n(1-r)} \times \frac{an(1-r)r^{n-1}}{1-r^n} = a^2 r^{n-1} = \left[ar^{\frac{n-1}{2}}\right]^2 = G^2$

**Problem 6:** The expenditure of 1,000 families is given as under:

Expenditure :      40 – 59    60 – 79    80 – 99    100 – 119    120 – 139

Number of families:    50                      --                      500                      --                      50

The median and mean of the distribution are both Rs. 87.50 respectively. Find the missing figures.

**Solution**

Let  $x$  and  $y$  be the missing figures for the balance classes 59.5 – 79.5 and 99.5 – 119.5, respectively. The data and necessary calculations shown in table:

| Class         | Frequency | Mid-x | f x      | LT cum.freq. |
|---------------|-----------|-------|----------|--------------|
| 39.5 – 49.5   | 50        | 49.5  | 2475     | 50           |
| 59.5 – 79.5   | $x$       | 69.5  | $69.5x$  | $50+x$       |
| 79.5 – 99.5   | 500       | 89.5  | 44750    | $550+x$      |
| 99.5 – 119.5  | $y$       | 109.5 | $109.5y$ | $550+x+y$    |
| 119.5 – 139.5 | 50        | 129.5 | 475      | $600+x+y$    |

$$N = 1000 \quad \sum fx = 53700 + 69.5x + 109.5y$$

Given, Mean = 87.50

$$\Rightarrow \text{mean} = \frac{1}{1000}(53700 + 69.5x + 109.5y) = 87.50$$

$$69.5x + 109.5y = 33800 \quad \text{-----} \quad (1)$$

Given, Median = 87.5.

Here median class is 79.5 – 99.5.

$$\text{Then, median} = 79.5 + \left( \frac{\frac{1000}{2} - (50 + x)}{500} \right) \times 20 = 87.50$$

$$\Rightarrow 300 = 50 + x$$

$$\Rightarrow x = 250$$

Then by (1), we get  $y = 150$ .

The missing frequencies are 250 and 150.

## 2.8. OGIVES (Cumulative frequency curves)

For a given frequency table, ogives are the graphs of cumulated frequencies. They can be used to determine how many data values lie above or below a particular value in the given frequency table. Consider the data given in the form of a grouped frequency table. A table consists of the lower bound of each class in the first column and the number of observations greater than each lower bound (greater than cumulative frequencies) in second column is called 'greater than cumulative frequency table'.

In the same way, a table consists of the upper bound of each class in the first column and the number of observations less than each upper bound (less than cumulative frequencies) in second column is called 'less than cumulative frequency table'.